# Field Effect Transistor - N-Channel, Enhancement Mode

# BS170, MMBF170

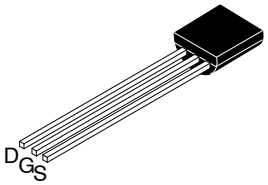
### General Description

These N-Channel enhancement mode field effect transistors are produced using **onsemi**'s proprietary, high cell density, DMOS technology. These products have been designed to minimize on-state resistance while provide rugged, reliable, and fast switching performance. They can be used in most applications requiring up to 500 mA DC. These products are particularly suited for low voltage, low current applications such as small servo motor control, power MOSFET gate drivers, and other switching applications.

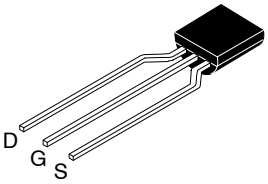
### Features

- High Density Cell Design for Low $R_{DS(ON)}$
- Voltage Controlled Small Signal Switch
- Rugged and Reliable
- High Saturation Current Capability
- These are Pb-Free Devices

**BS170**

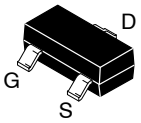



**TO-92 3 4.825x4.76**
**CASE 135AN**

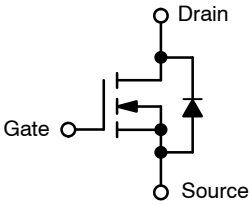



**TO-92 3 4.83x4.76 LEADFORMED**
**CASE 135AR**

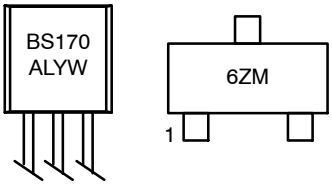
**MMBF170**




**SOT-23**
**CASE 318-08**




### MARKING DIAGRAM




| | |
|---|---|
| BS170, 6Z | = Device Code |
| A | = Assembly Plant Code |
| L | = Wafer Lot Number |
| YW | = Assembly Start Week |
| M | = Date Code |

### ORDERING INFORMATION

See detailed ordering and shipping information on page 8 of this data sheet.

NOTE: Some of the devices on this data sheet have been **DISCONTINUED**. Please refer to the table on page 8.

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## ABSOLUTE MAXIMUM RATINGS ($T_A$ = 25 °C unless otherwise noted)

| Symbol | Parameter | | BS170 | MMBF170 | Unit |
|---|---|---|---|---|---|
| $V_{DSS}$ | Drain-Source Voltage | | 60 | | V |
| $V_{DGR}$ | Drain-Gate Voltage ($R_{GS}$ ≤ 1 MΩ) | | 60 | | V |
| $V_{GSS}$ | Gate-Source Voltage | | ±20 | | V |
| $I_D$ | Drain Current | – Continuous | 500 | 500 | mA |
| | | – Pulsed | 1200 | 800 | |
| $T_J$, $T_{STG}$ | Operating and Storage Temperature Range | | – 55 to 150 | | °C |
| $T_L$ | Maximum Lead Temperature for Soldering Purposes, 1/16” from Case for 10 Seconds | | 300 | | °C |

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

## THERMAL CHARACTERISTICS ($T_A$ = 25 °C unless otherwise noted)

| Symbol | Parameter | BS170 | MMBF170 | Unit |
|---|---|---|---|---|
| $P_D$ | Maximum Power Dissipation<br>Derate above 25 °C | 830<br>6.6 | 300<br>2.4 | mW<br>mW/°C |
| $R_{\theta JA}$ | Thermal Resistance, Junction to Ambient | 150 | 417 | °C/W |

## ELECTRICAL CHARACTERISTICS ($T_A$ = 25 °C unless otherwise noted)

| Symbol | Parameter | Test Condition | Type | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|---|
| **OFF CHARACTERISTICS** | | | | | | | |
| $BV_{DSS}$ | Drain-Source Breakdown Voltage | $V_{GS}$ = 0 V, $I_D$ = 100 μA | All | 60 | – | – | V |
| $I_{DSS}$ | Zero Gate Voltage Drain Current | $V_{DS}$ = 25 V, $V_{GS}$ = 0 V | All | – | – | 0.5 | μA |
| $I_{GSSF}$ | Gate - Body Leakage, Forward | $V_{GS}$ = 15 V, $V_{DS}$ = 0 V | All | – | – | 10 | nA |
| **ON CHARACTERISTICS** (Note 1) | | | | | | | |
| $V_{GS(th)}$ | Gate Threshold Voltage | $V_{DS}$ = $V_{GS}$, $I_D$ = 1 mA | All | 0.8 | 2.1 | 3 | V |
| $R_{DS(ON)}$ | Static Drain-Source On-Resistance | $V_{GS}$ = 10 V, $I_D$ = 200 mA | All | – | 1.2 | 5 | Ω |
| $g_{FS}$ | Forward Transconductance | $V_{DS}$ = 10 V, $I_D$ = 200 mA | BS170 | – | 320 | – | mS |
| | | $V_{DS}$ ≥ 2 $V_{DS(on)}$, $I_D$ = 200 mA | MMBF170 | – | 320 | – | |
| **DYNAMIC CHARACTERISTICS** | | | | | | | |
| $C_{iss}$ | Input Capacitance | $V_{DS}$ = 10 V, $V_{GS}$ = 0 V, f = 1.0 MHz | All | – | 24 | 40 | pF |
| $C_{oss}$ | Output Capacitance | | All | – | 17 | 30 | pF |
| $C_{rss}$ | Reverse Transfer Capacitance | | All | – | 7 | 10 | pF |
| **SWITCHING CHARACTERISTICS** (Note 1) | | | | | | | |
| $t_{on}$ | Turn-On Time | $V_{DD}$ = 25 V, $I_D$ = 200 mA, $V_{GS}$ = 10 V, $R_{GEN}$ = 25 Ω | BS170 | – | – | 10 | ns |
| | | $V_{DD}$ = 25 V, $I_D$ = 500 mA, $V_{GS}$ = 10 V, $R_{GEN}$ = 50 Ω | MMBF170 | – | – | 10 | |
| $t_{off}$ | Turn-Off Time | $V_{DD}$ = 25 V, $I_D$ = 200 mA, $V_{GS}$ = 10 V, $R_{GEN}$ = 25 Ω | BS170 | – | – | 10 | ns |
| | | $V_{DD}$ = 25 V, $I_D$ = 500 mA, $V_{GS}$ = 10 V, $R_{GEN}$ = 50 Ω | MMBF170 | – | – | 10 | |

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

1. Pulse Test: Pulse Width ≤ 300 μs, Duty Cycle ≤ 2.0%.



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## TYPICAL ELECTRICAL CHARACTERISTICS

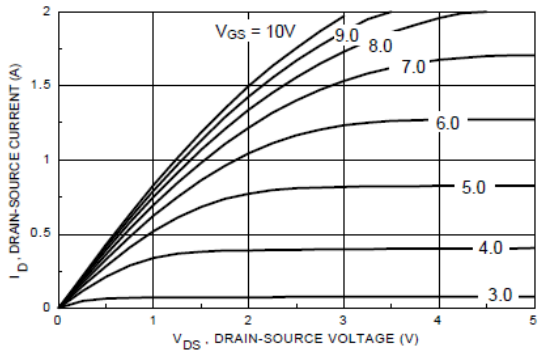


**Figure 1. On-Region Characteristics**

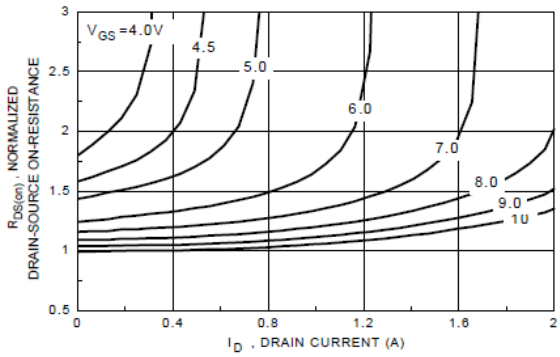


**Figure 2. On-Resistance Variation with Gate Voltage and Drain Current**

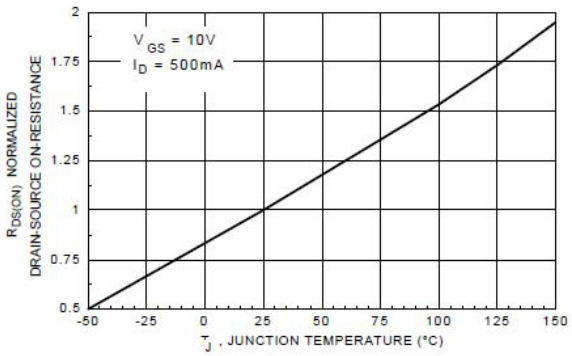


**Figure 3. On-Resistance Variation with Temperature**

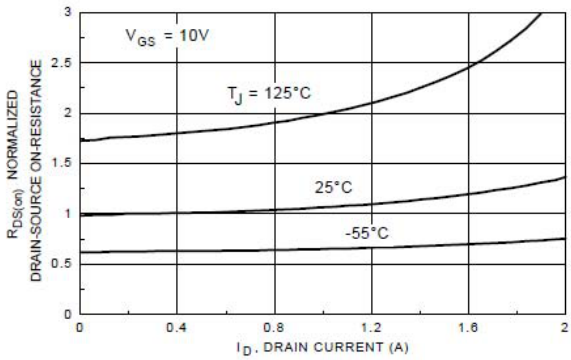


**Figure 4. On-Resistance Variation with Drain Current and Temperature**

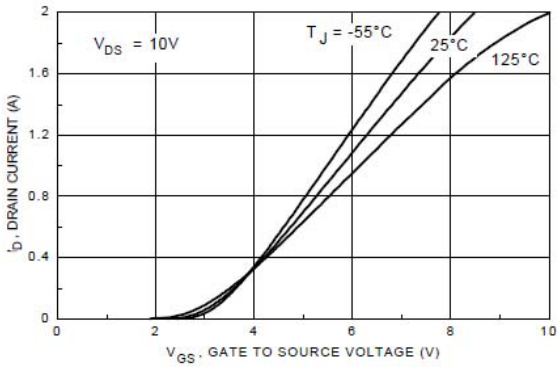


**Figure 5. Transfer Characteristics**

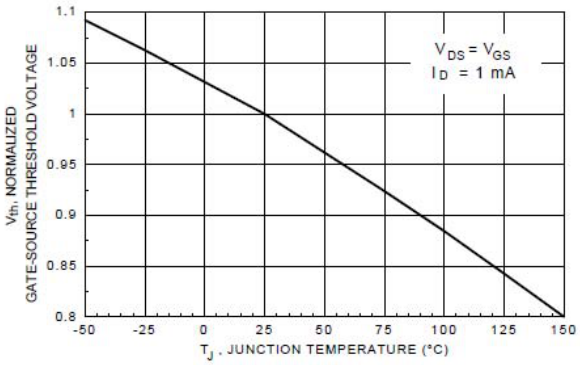


**Figure 6. Gate Threshold Variation with Temperature**



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## TYPICAL ELECTRICAL CHARACTERISTICS (CONTINUED)

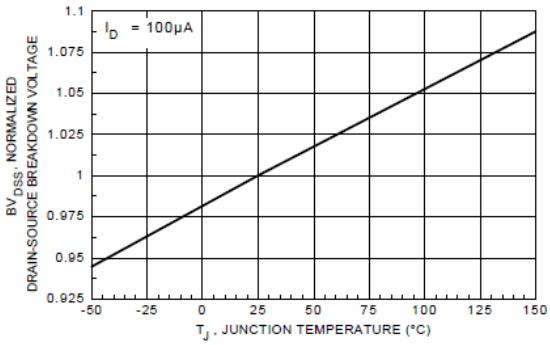


**Figure 7. Breakdown Voltage Variation with Temperature**

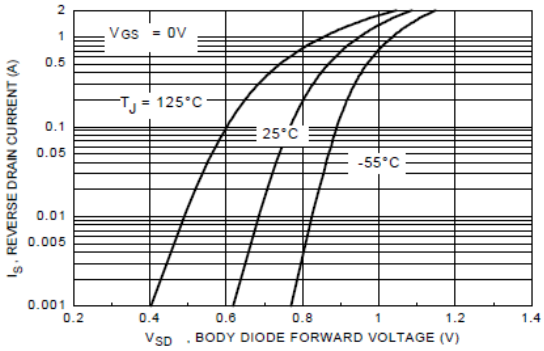


**Figure 8. Body Diode Forward Voltage Variation with Current and Temperature**

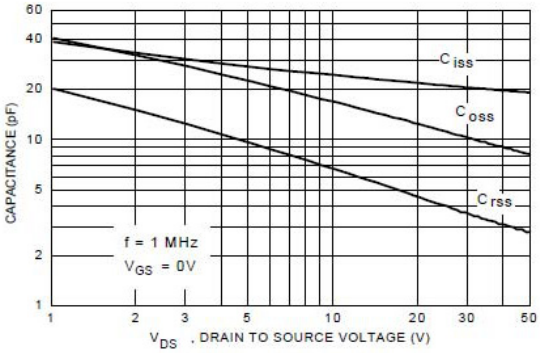


**Figure 9. Capacitance Characteristics**

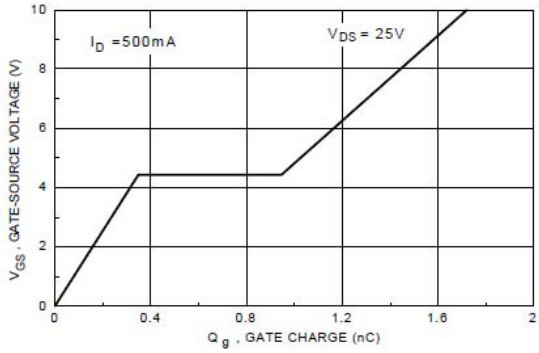


**Figure 10. Gate Charge Characteristics**

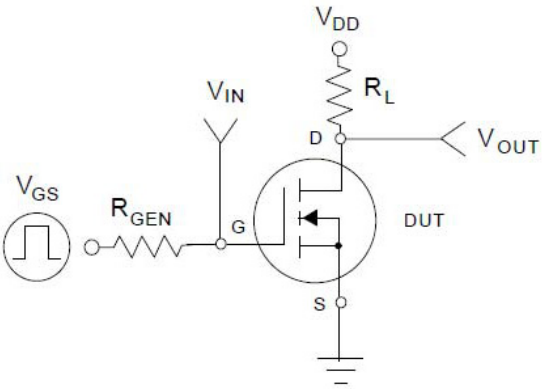


**Figure 11. Switching Test Circuit**

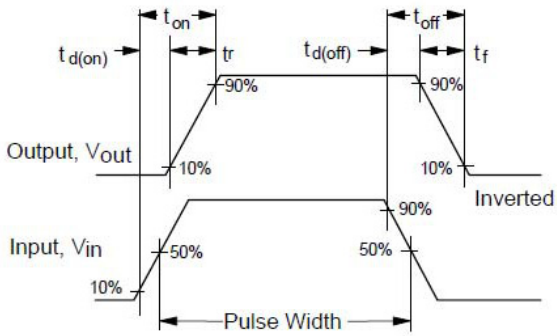


**Figure 12. Switching Waveforms**



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## TYPICAL ELECTRICAL CHARACTERISTICS (CONTINUED)

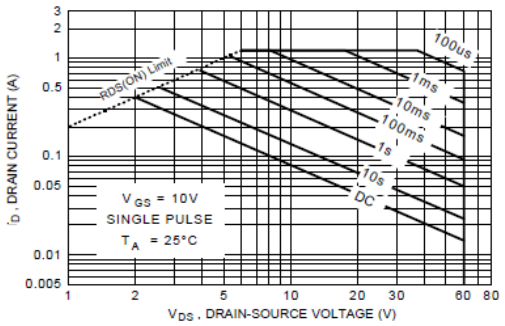


**Figure 13. BS170 Maximum Safe Operating Area**

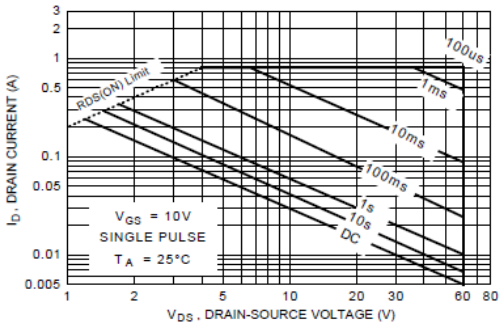


**Figure 14. MMBF170 Maximum Safe Operating Area**

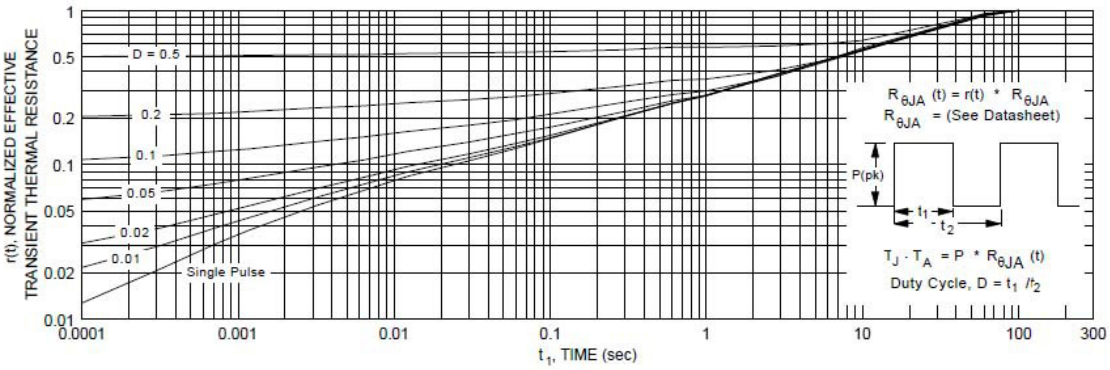


**Figure 15. TO-92, BS170 Transient Thermal Response Curve**

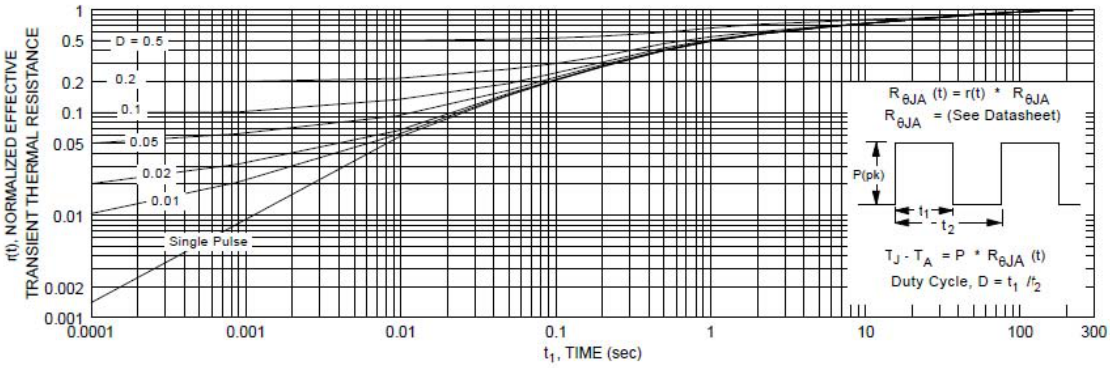


**Figure 16. SOT-23, MMBF170 Transient Thermal Response Curve**



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## ORDERING INFORMATION

| Part Number | Package | Lead Frame | Pin Array | Shipping† |
|---|---|---|---|---|
| BS170 | TO-92 (Pb-Free) | Straight | D G S | 10,000 Units / Bulk |
| BS170-D26Z | TO-92 (Pb-Free) | Forming | D G S | 2,000 / Tape & Reel |
| BS170-D27Z | TO-92 (Pb-Free) | Forming | D G S | 2,000 / Tape & Reel |
| MMBF170 | SOT-23 (Pb-Free) | | | 3,000 / Tape & Reel |

**DISCONTINUED** (Note 11)

| | | | | |
|---|---|---|---|---|
| BS170-D74Z | TO-92 (Pb-Free) | Forming | D G S | 2,000 / Ammo |
| BS170-D75Z | TO-92 (Pb-Free) | Forming | D G S | 2,000 / Ammo |

† For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

2. **DISCONTINUED:** These devices are not available. Please contact your **onsemi** representative for information. The most current information on these devices may be available on www.onsemi.com.



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## REVISION HISTORY

| Revision | Description of Changes | Date |
|---|---|---|
| 9 | BS170-D75Z OPN Marked as Discontinued. | 02/12/2026 |

This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.



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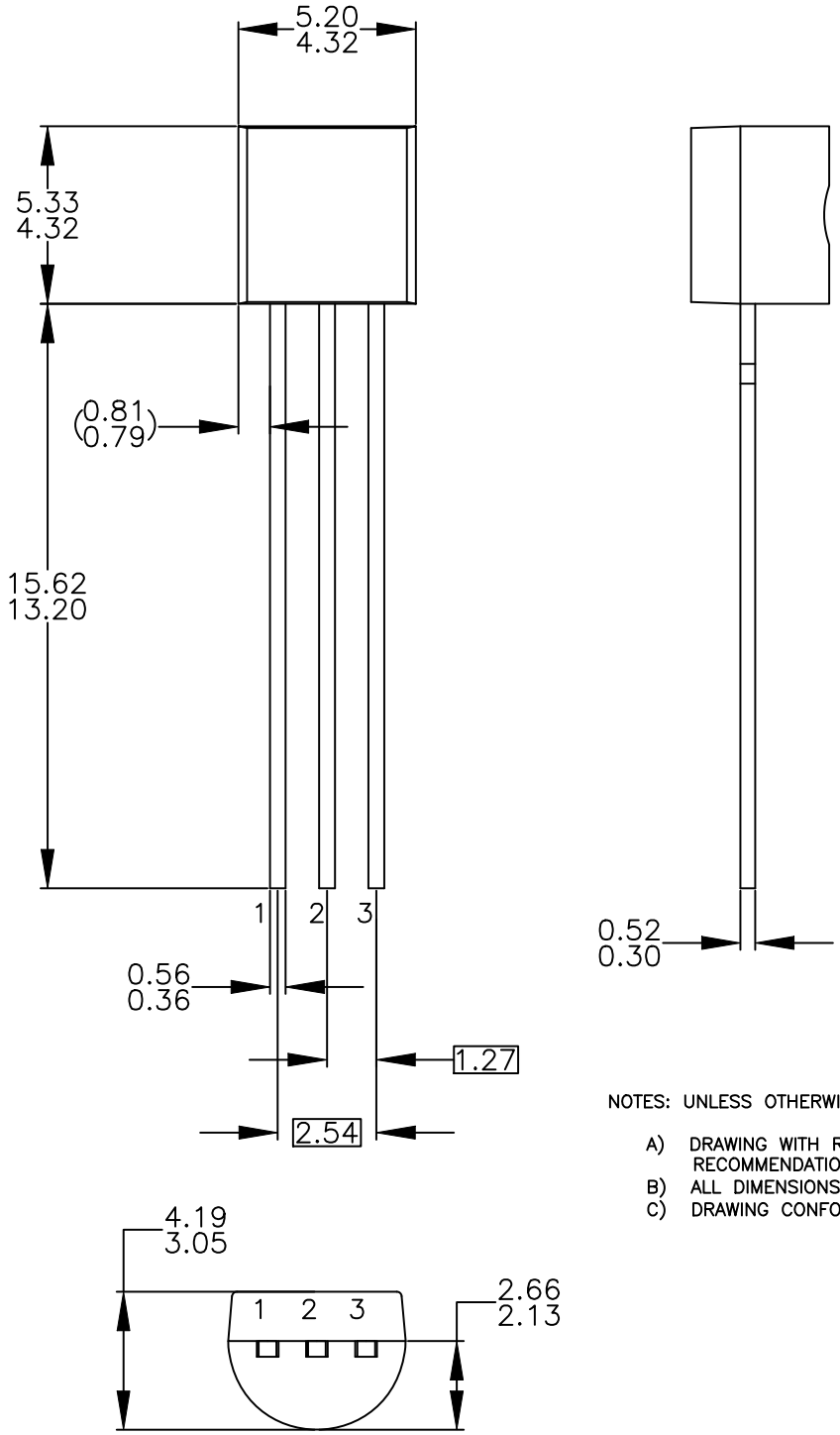
**TO−92 3 4.825x4.76**
CASE 135AN
ISSUE O

DATE 31 JUL 2016




NOTES: UNLESS OTHERWISE SPECIFIED

A) DRAWING WITH REFERENCE TO JEDEC TO−92 RECOMMENDATIONS.
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C) DRAWING CONFORMS TO ASME Y14.5M−2009.

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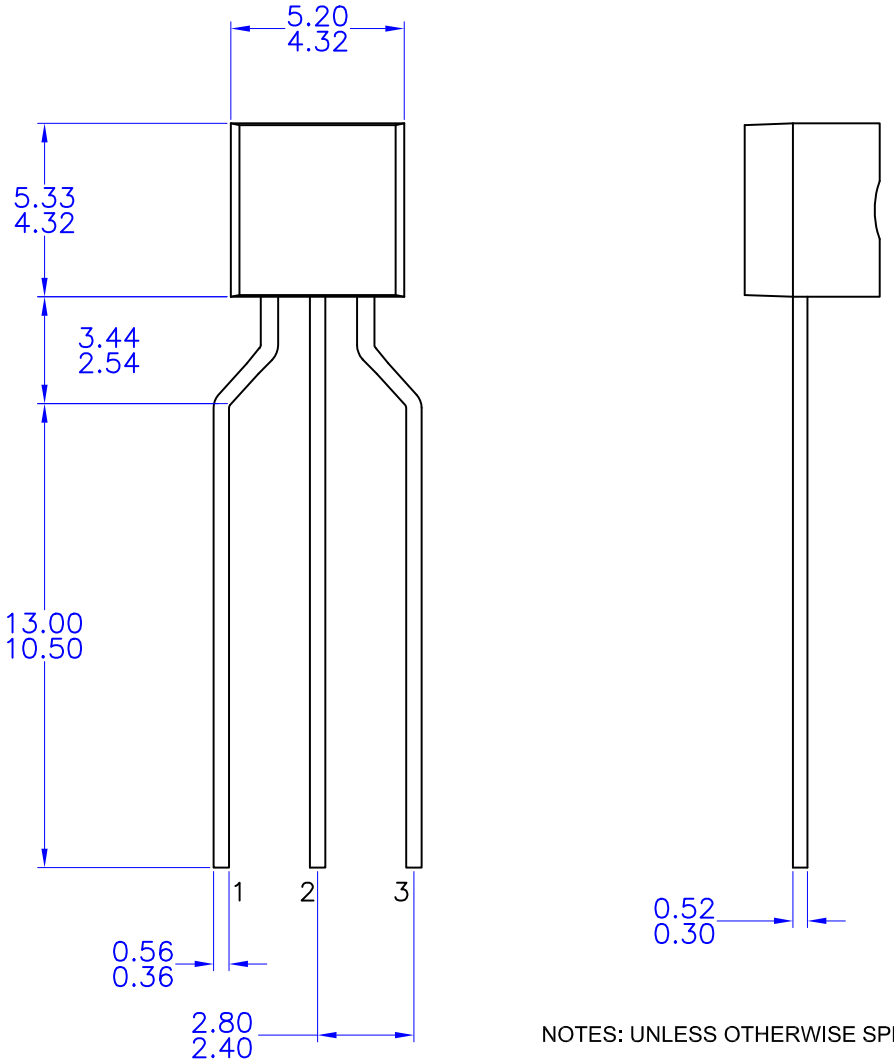
**TO−92 3 4.83x4.76 LEADFORMED**
CASE 135AR
ISSUE O

DATE 30 SEP 2016

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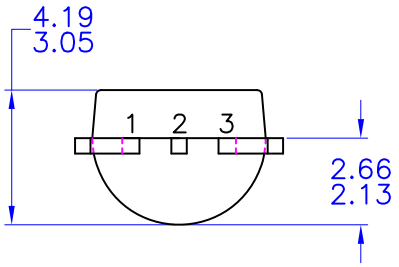


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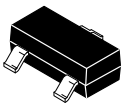
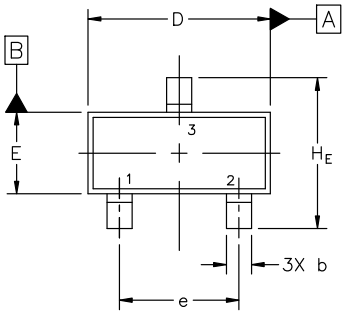


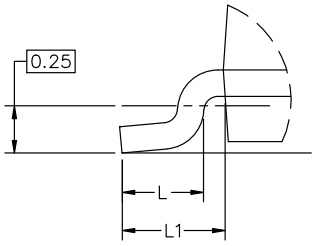
SCALE 4:1

**SOT−23 (TO−236) 2.90x1.30x1.00 1.90P**
CASE 318
ISSUE AU

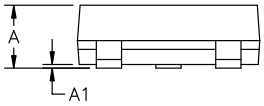
DATE 14 AUG 2024



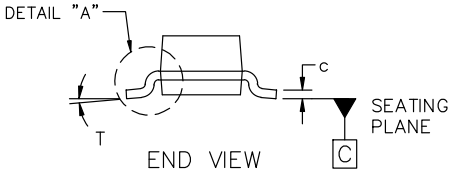

TOP VIEW




DETAIL "A"
Scale 3:1




SIDE VIEW

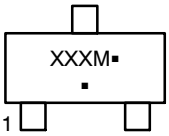



END VIEW

| MILLIMETERS | | | |
|---|---|---|---|
| DIM | MIN | NOM | MAX |
| A | 0.89 | 1.00 | 1.11 |
| A1 | 0.01 | 0.06 | 0.10 |
| b | 0.37 | 0.44 | 0.50 |
| c | 0.08 | 0.14 | 0.20 |
| D | 2.80 | 2.90 | 3.04 |
| E | 1.20 | 1.30 | 1.40 |
| e | 1.78 | 1.90 | 2.04 |
| L | 0.30 | 0.43 | 0.55 |
| L1 | 0.35 | 0.54 | 0.69 |
| $H_E$ | 2.10 | 2.40 | 2.64 |
| T | 0° | −−− | 10° |

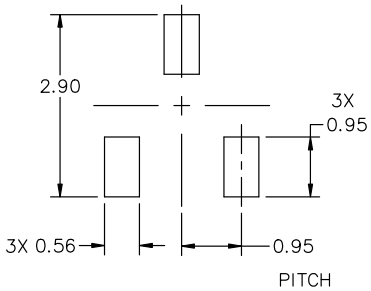
NOTES:
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2018.
2. CONTROLLING DIMENSIONS: MILLIMETERS.
3. MAXIMUM LEAD THICKNESS INCLUDES LEAD FINISH. MINIMUM LEAD THICKNESS IS THE MINIMUM THICKNESS OF THE BASE MATERIAL.
4. DIMENSIONS D AND E DO NOT INCLUDE MOLD FLASH, PROTRUSIONS, OR GATE BURRS.

**GENERIC MARKING DIAGRAM***




XXX = Specific Device Code
M = Date Code
▪ = Pb−Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.




RECOMMENDED
MOUNTING FOOTPRINT

* For additional information on our Pb−Free strategy and soldering details, please download the onsemi Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

**STYLES ON PAGE 2**

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| **DESCRIPTION:** | **SOT−23 (TO−236) 2.90x1.30x1.00 1.90P** | **PAGE 1 OF 2** |

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**SOT−23 (TO−236) 2.90x1.30x1.00 1.90P**
CASE 318
ISSUE AU

DATE 14 AUG 2024

STYLE 1 THRU 5:
CANCELLED

STYLE 6:
PIN 1. BASE
2. EMITTER
3. COLLECTOR

STYLE 7:
PIN 1. EMITTER
2. BASE
3. COLLECTOR

STYLE 8:
PIN 1. ANODE
2. NO CONNECTION
3. CATHODE

STYLE 9:
PIN 1. ANODE
2. ANODE
3. CATHODE

STYLE 10:
PIN 1. DRAIN
2. SOURCE
3. GATE

STYLE 11:
PIN 1. ANODE
2. CATHODE
3. CATHODE−ANODE

STYLE 12:
PIN 1. CATHODE
2. CATHODE
3. ANODE

STYLE 13:
PIN 1. SOURCE
2. DRAIN
3. GATE

STYLE 14:
PIN 1. CATHODE
2. GATE
3. ANODE

STYLE 15:
PIN 1. GATE
2. CATHODE
3. ANODE

STYLE 16:
PIN 1. ANODE
2. CATHODE
3. CATHODE

STYLE 17:
PIN 1. NO CONNECTION
2. ANODE
3. CATHODE

STYLE 18:
PIN 1. NO CONNECTION
2. CATHODE
3. ANODE

STYLE 19:
PIN 1. CATHODE
2. ANODE
3. CATHODE−ANODE

STYLE 20:
PIN 1. CATHODE
2. ANODE
3. GATE

STYLE 21:
PIN 1. GATE
2. SOURCE
3. DRAIN

STYLE 22:
PIN 1. RETURN
2. OUTPUT
3. INPUT

STYLE 23:
PIN 1. ANODE
2. ANODE
3. CATHODE

STYLE 24:
PIN 1. GATE
2. DRAIN
3. SOURCE

STYLE 25:
PIN 1. ANODE
2. CATHODE
3. GATE

STYLE 26:
PIN 1. CATHODE
2. ANODE
3. NO CONNECTION

STYLE 27:
PIN 1. CATHODE
2. CATHODE
3. CATHODE

STYLE 28:
PIN 1. ANODE
2. ANODE
3. ANODE




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